

LeadOnSpot - AI Engineer Intern Assignment

Assignment: Build an AI Lead Qualification Agent

Overview: Build a REST API that accepts lead information and returns AI-generated lead qualification insights.

Requirements:

- Backend: FastAPI (preferred) or Node.js
- LLM: OpenAI, Claude, or Gemini
- Endpoint: POST /api/v1/lead/qualify

Input fields include name, email, phone, company, designation, industry, employees, country, source and message.

The API should return structured JSON containing: lead_score (0-100), priority (HOT/WARM/COLD), buying_intent, intent, company_size, estimated_deal_size, pain_points, recommended_next_action, sales_summary and confidence score.

Functional requirements: request validation, structured JSON output, graceful error handling, logging, prompt engineering for deterministic JSON responses.

Bonus: cost tracking, prompt versioning, caching, Docker, unit tests, evaluation dataset.

Deliverables: GitHub repository, README, API documentation, sample requests/responses, optional Postman collection and Docker support.

Evaluation: Code Quality 25%, Prompt Engineering 20%, AI Output 20%, API Design 10%, Error Handling 10%, Documentation 5%, Testing 5%, Bonus 5%.